

More on Next Diary

It is important to consider which elements in our proof are arbitrary & which are not.

And what is the order of quantifiers. e.g. $\forall x \exists y$ means there is ^{for every x , a unique y .} a single y for every x . If ~~there~~ ^{can} y depend on x .
 & $\exists! y \forall x$ means y cannot depend on x .
 & there is a single y that works for every x .

If I have to prove $\exists! x \forall y$ then I can choose any specific value of x but if I have to prove $\forall y \exists! x$ then I have to let y be arbitrary ~~not specific~~ so every y , there is a unique x that could depend on y but in first case, there is one x for every y .

For a universal statement. One contradiction is enough to make it false. Sometimes it's hard to get anywhere when we try to find a contradiction using arbitrary elements.

e.g. $\forall x [x \in A \rightarrow x \in B]$

If we can show an eg. such that $x \in A$ & $x \notin B$ that will be enough to contradict $\forall$ quantifier. That can be a specific example. (see ex 10 of Lec 3.6)

Always try to identify possible contradictions when doing proof by contradiction. Sometimes it's not clear when will contradiction occur but identifying possibilities will make it clear. e.g. suppose we have

$\frac{x}{n+1} < y$. & we have to prove that $y > 1$

& we suppose that $y \leq 1$ to show contradiction. Then + we are possible contradiction:

$$y > 1, y = 1, \frac{x}{n+1} > y, \frac{x}{n+1} = y$$

look out for specific properties of something. e.g. suppose we are given $x = \phi$. We can use the property that $\phi \subseteq A$ for every set A . Also $\phi \subseteq \phi$. $\phi \in \{\phi\}$. So try to think of properties.

if we have $\forall x [x \in A \rightarrow x \in B]$ & we know that $\forall x (x \notin A)$ we cannot conclude that $x \in B$ but we can conclude that $\forall x [x \in A \rightarrow x \in B]$ is true - Vacuous Proof.

e.g. $\forall x [x \in \phi \rightarrow x \in A]$ we know $x \in \phi$ is F so overall implication is true, which translates to $\phi \subseteq A$.

if we have $x \in A$ & $A \subseteq B$ where A is arbitrary then we can conclude $x \in U$ & $x \in \cap F$. As A is arbitrary $\forall A$ @ & $\exists A$ both can be concluded as $\forall A$ also means $\exists A$.

But if we have a specific $A \subseteq B$ & $x \in A$ then we can only conclude $x \in U$ & not $x \in \cap F$.

Any set defined like $A = \{x \mid x \mid 2\}$ means $\forall x [x \in A \leftrightarrow x \mid 2]$

Quantification order is very important. Ex: If we have

$p \rightarrow (q \rightarrow r)$ we can use PE of q & then prove r

But if we have $\forall x (\exists y P(x, y) \rightarrow Q(x))$ its clearly different

from $\forall x \forall y (P(x, y) \rightarrow Q(x))$.

So when quantifiers are nested
is important.